10.2 The Stacked-Ball Effect

WORKSHEET

The question

STEP II 2010, Question 10

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/10-s2-q10

Hints

Try the question on your own first. If you get stuck, read the hints *one at a time*, and go back to the question after each one. Hints marked *Pause and try* are the key steps. There is space after each hint for your working.

Ask yourself From Section 10.1: if a mass M moving at speed u hits a stationary mass m elastically, what is the speed of m afterwards?

STAGE 1 The speed limit

Covers: Part (i)

Ask yourself Use conservation of momentum and the elastic collision (or the result for a moving mass hitting a stationary one) to find the speed of A in terms of b and u.

Pause and try Write $\frac{2b}{b+1}$ as 2 minus something positive. What happens as $b \to \infty$?

STAGE 2 Passing it along a chain

Covers: Part (ii)

Ask yourself Each particle is λ times as heavy as the next one along. So each collision is Part (i) with $b = \lambda$. By what factor is the speed multiplied at each collision?

Pause and try After n collisions the speed is u times the nth power of that factor. Is the factor bigger or smaller than 1 when $\lambda > 1$?

STAGE 3 Twenty times as fast

Covers: the case $\lambda = 4$

Ask yourself With $\lambda = 4$ you need $1.6^n > 20$. Take logarithms to base 10, writing $1.6 = 16/10$ and $20 = 2 \times 10$.

What's really going on

The basketball and the tennis ball

Let the two balls fall together from height H, reaching the floor at speed $U = \sqrt{2gH}$. Suppose everything is perfectly elastic and the basketball is much heavier. It bounces off the floor with speed U upwards, and immediately meets the tennis ball, still moving down at speed U. In the basketball's frame, the tennis ball approaches at $2U$ and, bouncing off a very heavy object, leaves at $2U$. Back in the ground frame, that is $2U + U = 3U$ upwards (Figure 10.2.1). Since height is proportional to the square of the launch speed, the tennis ball rises to $9H$.

Real balls are not perfectly elastic. With coefficient of restitution e and mass ratio M/m, the tennis ball leaves at speed $\frac{Me(2+e)-m}{M+m}U$. STEP I 2014 Q10, in the exercises, uses a measured height to work backwards to the mass ratio.

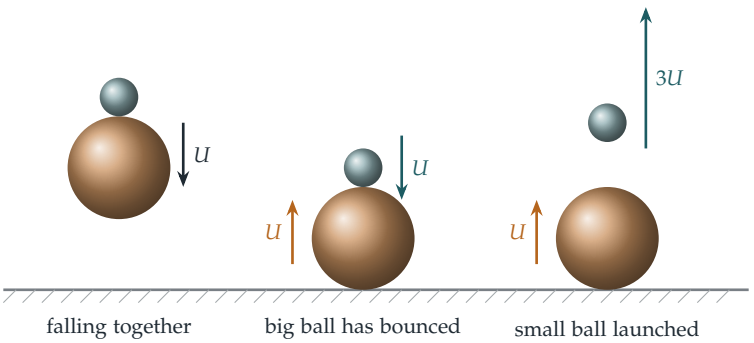


Figure 10.2.1 The stacked-ball effect, for perfectly elastic balls with the lower one much heavier. The big ball bounces up at speed U and meets the small ball coming down at U. Relative to the big ball, the small one arrives at $2U$ and leaves at $2U$, so it leaves the ground frame at $3U$, reaching nine times the original height.

Gravitational slingshots

A space probe flying past a planet does not touch it, but gravity acts like a very soft elastic collision with a very heavy object. In the planet's frame, the probe swings round and leaves with the same speed it arrived with. If the planet is moving towards the probe at speed V, then in the Sun's frame the probe can gain up to $2V$, exactly as the tennis ball does. The two *Voyager* probes, launched in 1977, used a rare alignment of the outer planets to pass from Jupiter to Saturn (and, for *Voyager 2*, on to Uranus and Neptune), gaining speed at each encounter. Without these gravity assists, the journey to Neptune would have taken about thirty years instead of twelve. The energy comes from the planet, which slows down by an immeasurably small amount.

Superballs and supernovae

In 1968 W. R. Mellen described in the *American Journal of Physics* a "superball rebound projectile" made by stacking several balls of decreasing size, and noted that the top ball can be launched at many times the impact speed. Astrophysicists have pointed out a loose analogy with a core-collapse supernova: the infalling outer layers of a star rebound off the stiff, dense core, and a shock wave carries momentum outwards into ever less dense material, accelerating it to enormous speeds. Demonstrations of the stacked-ball effect are used to explain the idea to students, though real supernovae involve much more physics, including neutrinos, than bouncing balls.

Try this next

1. **The real thing.** Try STEP I 2014, Question 10 (step.maths.org/questions/14-s1-q10): the stacked-ball problem with real numbers and imperfect bounces.

2. **An inelastic chain.** Try STEP I 2017, Question 10 (step.maths.org/questions/17-s1-q10). It sends a collision along a chain of particles whose masses grow geometrically. When is each particle involved in exactly two collisions?

3. **Three balls.** Three perfectly elastic balls are stacked, each much lighter than the one below. Show that the top ball leaves at $7U$ and rises to $49H$. With k balls, show that the top one leaves at $(2^k - 1)U$.

4. **Where does the energy come from?** In part (ii), show that the total kinetic energy is conserved even though A's speed is unbounded. What happens to the heavy particles?

Further reading

- **David Morin, *Introduction to Classical Mechanics*** (Cambridge University Press), Chapter 5, which includes the stacked-ball problem.
- **W. R. Mellen, "Superball rebound projectiles"**, *American Journal of Physics* 36 (1968).

Short and fun.

- **Gary Flandro, "Fast reconnaissance missions to the outer solar system utilizing energy derived from the gravitational field of Jupiter"**, *Astronautica Acta* 12 (1966). The paper that proposed the Voyager "grand tour".

The full guided solution is in Section 10.2:
https://beyondstep.keithmcnulty.org/ch10/stacked-balls.html